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United States Patent	12396886
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Sacks; Zachary Shane et al.

Automated capsulotomy

Abstract

A system (20) includes a radiation source (48) and a controller (44). The controller is configured to define a treatment zone (88) on a capsule (86) of an eye (25) of a subject (22), and to form an opening (96) in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions (94) within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process, acquiring an image (98) of at least part of the capsule, designating one of the target regions based on the acquired image, and causing the radiation source to irradiate the designated target region. Other embodiments are also described.

Inventors:	Sacks; Zachary Shane (Modiin, IL), Belkin; Michael (Givat Shmu'el, IL)
Applicant:	BELKIN VISION LTD. (Yavne, IL)
Family ID:	1000008781031
Assignee:	Belkin Vision Ltd. (Yavne, IL)
Appl. No.:	17/926633
Filed (or PCT Filed):	May 16, 2021
PCT No.:	PCT/IB2021/054187
PCT Pub. No.:	WO2022/018525
PCT Pub. Date:	January 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230201034 A1	Jun. 29, 2023

Related U.S. Application Data

us-provisional-application US 63053650 20200719

Publication Classification

Int. Cl.: A61F9/008 (20060101)

U.S. Cl.:

CPC A61F9/008 (20130101); A61F2009/00844 (20130101); A61F2009/0087 (20130101); A61F2009/00889 (20130101); A61F2009/00897 (20130101)

Field of Classification Search

CPC: A61F (9/008); A61F (2009/00844); A61F (2009/0087); A61F (2009/00889); A61F (2009/00897)

USPC: 606/4

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Primary Examiner: Roane; Aaron F

Attorney, Agent or Firm: Meitar Patents Ltd.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of U.S. Provisional Application 63/053,650, entitled “Automatic posterior capsulotomy and combination device,” filed Jul. 19, 2020, whose disclosure is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention is related to ophthalmologic procedures, especially capsulotomy procedures.

BACKGROUND

(2) In some cases, following the replacement of the natural lens of an eye, opacification of the posterior capsule of the eye occurs. In such cases, a posterior capsulotomy may be the treatment of choice.

(3) U.S. Pat. No. 8,465,478 describes systems, apparatus, and methods for developing laser systems that can create a precise predetermined jigsaw capsulotomy. The systems, apparatus, and methods further provide laser systems that can use a single laser as a therapeutic laser and as laser radar and that reduce the patient-to-patient variability and doctor-to-doctor variability associated with hand-held apparatus for performing capsulorhexis and capsulotomies. There is further described a precise predetermined jigsaw shot pattern and shaped capsulotomy that is based at least in part on the shape of an IOL and in particular an accommodating IOL.

(4) U.S. Pat. No. 8,845,625 describes systems and methods for cataract intervention. In one embodiment a system comprises a laser source configured to produce a treatment beam comprising a plurality of laser pulses; an integrated optical system comprising an imaging assembly operatively coupled to a treatment laser delivery assembly such that they share at least one common optical element, the integrated optical system being configured to acquire image information pertinent to one or more targeted tissue structures and direct the treatment beam in a 3-dimensional pattern to cause breakdown in at least one of the targeted tissue structures; and a controller operatively coupled to the laser source and integrated optical system, and configured to adjust the laser beam and treatment pattern based upon the image information, and distinguish two or more anatomical structures of the eye based at least in part upon a robust least squares fit analysis of the image information.

(5) U.S. Pat. No. 10,143,590 describes a method and apparatus for performing a laser-assisted posterior capsulotomy and for performing laser eye surgery on an eye having a penetrated cornea. A method for performing a posterior capsulotomy includes injecting fluid between the lens posterior capsule and the anterior hyaloids membrane to separate the lens posterior capsule and the anterior hyaloids membrane. With the lens posterior capsule separated from the anterior hyaloids membrane, a posterior capsulotomy is performed on the lens posterior capsule by using a laser to incise the lens posterior capsule.

(6) U.S. Pat. No. 10,849,789 describes an ophthalmic measurement and laser surgery system,

including: a laser source; a corneal topography subsystem; an axis determining subsystem; a ranging subsystem comprising an Optical Coherence Tomographer (OCT); and a refractive index determining subsystem. All of the subsystems are under the operative control of a controller. The controller is configured to: operate the corneal topography subsystem to obtain corneal surface information; operate the axis determining subsystem to identify one or more ophthalmic axes of the eye; operate the OCT to sequentially scan the eye in a plurality of OCT scan patterns, the plurality of scan patterns configured to determine an axial length of the eye; operate the refractive index determining subsystem so to determine an index of refraction of one or more ophthalmic tissues, wherein at least one of the corneal surface information, ophthalmic axis information, and axial length is modified based on the determined index of refraction.

SUMMARY OF THE INVENTION

(7) There is provided, in accordance with some embodiments of the present invention, a system, including a radiation source and a controller. The controller is configured to define a treatment zone on a capsule of an eye of a subject, and to form an opening in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing the radiation source to irradiate the designated target region.

(8) In some embodiments, the capsule is a posterior capsule.

(9) In some embodiments, the controller is configured to define the treatment zone by:

(10) based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye, and

(11) defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.

(12) In some embodiments, the controller is configured to define the treatment zone by:

(13) based on at least one initial image of the eye, identifying an edge of an iris of the eye, and

(14) defining the treatment zone such that the treatment zone lies entirely within the edge.

(15) In some embodiments, the controller is configured to define the treatment zone by:

(16) based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye,

(17) in response to the features, calculating an estimated position of the IOL, and

(18) defining the treatment zone in response to the estimated position.

(19) In some embodiments, during at least one of the iterations, designating the target region includes:

(20) based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and

(21) in response to identifying the tissue, redesignating the last-designated one of the target regions.

(22) In some embodiments, during at least one of the iterations, designating the target region includes:

(23) based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and

(24) designating the target region at a predefined distance from the identified portion of the perimeter.

(25) In some embodiments, during at least one of the iterations, designating the target region includes:

(26) based on the image, identifying a fold within the treatment zone, and

(27) in response to identifying the fold, designating the target region such that the target region overlaps the fold.

(28) In some embodiments, during at least one of the iterations, designating the target region includes:

- (29) based on the acquired image, ascertaining that a perimeter of the opening is stable, and
- (30) in response to the ascertaining, designating the target region.
- (31) In some embodiments,
- (32) the controller is configured to form the opening subsequently to a designation of a sequence of tentative target regions, and
- (33) during at least one of the iterations, designating the target region includes: based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.
- (34) In some embodiments, during at least one other one of the iterations, designating the target region includes:
- (35) based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and
- (36) in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.
- (37) In some embodiments, during at least one other one of the iterations, designating the target region includes:
- (38) based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and
- (39) in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the target region by applying an offset to a location of the next one of the tentative target regions.
- (40) In some embodiments, the controller is further configured to:
- (41) define a target boundary on the capsule, and
- (42) terminate the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.
- (43) In some embodiments, the controller is configured to define the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.
- (44) There is further provided, in accordance with some embodiments of the present invention, a method, including defining a treatment zone on a capsule of an eye of a subject and, subsequently to defining the treatment zone, forming an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing a radiation source to irradiate the designated target region.
- (45) There is further provided, in accordance with some embodiments of the present invention, a computer software product including a tangible non-transitory computer-readable medium in which program instructions are stored. The instructions, when read by a controller, cause the controller to define a treatment zone on a capsule of an eye of a subject, and to form an opening in the capsule, subsequently to defining the treatment zone, by irradiating multiple target regions within the treatment zone in an iterative process. The iterative process includes, during each one of multiple iterations of the process, acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions, and causing the radiation source to irradiate the designated target region.
- (46) The present invention will be more fully understood from the following detailed description of embodiments thereof, taken together with the drawings, in which:
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of a system for performing a capsulotomy on a capsule of an eye of a patient, in accordance with some embodiments of the present invention;
- (2) FIG. 2 is a schematic illustration of a capsulotomy device, in accordance with some embodiments of the present invention;
- (3) FIG. 3 is a schematic illustration of a method for defining a treatment zone, in accordance with some embodiments of the present invention;
- (4) FIG. 4 is a schematic illustration of the formation of an opening in a capsule, in accordance with some embodiments of the present invention;
- (5) FIGS. 5-6 are schematic illustrations of techniques for designating target regions, in accordance with some embodiments of the present invention;
- (6) FIG. 7 is a schematic illustration of a technique for designating target regions based on a predesignated sequence of tentative target regions, in accordance with some embodiments of the present invention; and
- (7) FIG. 8 is a flow diagram for an example iterative process for forming an opening in a capsule of an eye, in accordance with some embodiments of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Overview

- (8) Conventional capsulotomy procedures require a specialist physician who has undergone extensive training. Unfortunately, however, such physicians may be in short supply.
- (9) To address this challenge, embodiments of the present invention provide an automated capsulotomy system, which allows even a physician who has undergone a lesser amount of training to perform the procedure. The system comprises a radiation source, a camera, a controller, and suitable optics. Based on images acquired by the camera, the controller controls the radiation source and optics so as to form an opening in the posterior capsule by irradiating the capsule.
- (10) Typically, the controller first defines a treatment zone, which includes at least some of the opacified portion of the posterior capsule. Next, the controller designates a first target region, typically at the center of the treatment zone, and then irradiates the first target region. After confirming, from an image of the capsule, that an opening was initialized at the first target region, the controller designates a second target region at a suitable distance from the perimeter of the opening, and then irradiates the second target region. After confirming, from another image of the capsule, that the opening was expanded, the controller designates and then irradiates a third target region. In this manner, the controller uses image processing to optimize the designation of the target regions, thus facilitating expanding the opening more efficiently, i.e., using fewer radiation beams. Upon ascertaining, from another image, that the opening has been sufficiently enlarged, the controller terminates the procedure.
- (11) Typically, the controller is configured to identify, by image processing, any folds in the posterior capsule. In response to identifying a fold, a target region overlapping the fold may be designated. Advantageously, by targeting folds in the capsule, the opening may be expanded more efficiently.
- (12) In some embodiments, a sequence of tentative target regions is designated prior to any irradiation of the capsule. Subsequently, the controller iterates through the tentative target regions. For each tentative target region, based on the proximity of the tentative target region to the perimeter of the opening, the controller decides whether to irradiate the tentative target region, skip the tentative target region, or irradiate a new target region that is offset from the tentative target region.
- (13) In addition to performing an automated capsulotomy of a posterior capsule as described

herein, the controller may form slit-shaped openings in the anterior capsule of the eye. Alternatively or additionally, the controller may use any of the techniques described herein to form a larger opening in the anterior capsule.

System Description

(14) Reference is initially made to FIG. 1, which is a schematic illustration of a system 20, comprising a capsulotomy device 21, for performing a capsulotomy on a capsule of an eye 25 of a patient 22, in accordance with some embodiments of the present invention. Reference is further made to FIG. 2, which is a schematic illustration of capsulotomy device 21, in accordance with some embodiments of the present invention.

(15) Capsulotomy device 21 comprises an optical unit 30 and a controller 44. Optical unit 30 comprises one or more beam-directing elements, comprising, for example, one or more galvo mirrors 50, which may be referred to collectively as a “galvo scanner,” and/or a beam combiner 56. Optical unit 30 further comprises a radiation source 48, which is configured to irradiate the capsule of eye 25 with one or more treatment beams 52 by emitting the treatment beams toward the beam-directing elements such that the beams are directed by the beam-directing elements toward the capsule. Typically, radiation source 48 is further configured to irradiate the capsule with a pair of aiming beams 53, which are configured to overlap one another only when the distance of the optical unit from the capsule and the positioning of the beam-directing elements provide a desired treatment spot size on the capsule. Thus, aiming beams 53 may be used to verify the spot size of each treatment beam prior to emitting the treatment beam, as further described below with reference to FIG. 8.

(16) In some embodiments, optical unit 30 further comprises one or more other optics. For example, the optical unit may comprise a beam expander 70, which expands and then re-collimates treatment beams 52. In such embodiments, typically, optical unit 30 further comprises a focusing module (comprising, for example, an F-theta lens or another type of lens), configured to focus the treatment beams.

(17) More specifically, before the emission of each treatment beam 52 from radiation source 48, or while the beam is being emitted, controller 44 aims the beam-directing elements at a target region of the capsule such that the beam is directed, by the beam-directing elements, toward the target region. For example, the beam may be deflected by galvo mirrors 50 through beam expander 70 toward beam combiner 56, and then deflected by the beam combiner through focusing module 72, such that the beam impinges on the target region with the desired spot size. (Since each treatment beam impinges on the capsule with a non-infinitesimal spot size, the present application generally describes each beam as impinging on a “region” of the capsule, rather than impinging at a “point.”) The beam thus follows a path 92, which extends from the most downstream of the beam-directing elements—such as focusing module 72—to the target region.

(18) Typically, the radiation source comprises a laser, such as an Nd:YAG laser. (Examples of off-the-shelf products incorporating an Nd:YAG laser include Optimis II by Quantel Medical of Cournon-d'Auvergne, France, and Selecta Duet by Lumenis of Yokneam, Israel.) The laser may be modified to include an attenuator, an energy meter, and/or a mechanical shutter. Alternatively or additionally to a laser, the radiation source may comprise any other suitable emitter of radiation.

(19) In some embodiments, the treatment beams include visible light. Alternatively or additionally, the treatment beams may include non-visible electromagnetic radiation, such as microwave radiation, infrared radiation, X-ray radiation, or gamma radiation. Typically, the wavelength of the treatment beams is between 400 and 1400 nm, e.g., 532 nm or 1064 nm.

(20) Typically, the spatial profile of each treatment beam 52 on the capsule is approximately circular, e.g., with a spot size less than 22 μm . Alternatively, the spatial profile of each treatment beam 52 may be elliptical, square, or of any other suitable shape. In general, the size and shape of the treatment beam spots are chosen so as to deliver sufficient energy for photodisruption of the capsule.

(21) Typically, the aiming beams include visible light, e.g., with a wavelength of between 600 and 700 nm.

(22) Optical unit **30** further comprises a camera **54**, which is used by controller **44** to acquire images of the eye. As shown in FIG. 2, camera **54** is typically aligned, at least approximately, with path **92**; for example, the angle between path **92** and a hypothetical line extending from eye **25** to the camera may be less than 15 degrees. In some embodiments, the camera is positioned behind beam combiner **56**, such that the camera receives light via the beam combiner. In other embodiments, the camera is offset from the beam combiner.

(23) At the start of the procedure, camera **54** acquires an initial image of the capsule, or at least a portion thereof. Based on the image, controller **44** defines a treatment zone on the capsule, as further described below with reference to FIG. 3. Subsequently, during the procedure, camera **54** acquires multiple images of the capsule, or at least a portion thereof, at a relatively high frequency. Controller **44** processes these images and, in response thereto, designates target regions within the treatment zone for irradiation, as further described below with reference to the subsequent figures.

(24) In general, camera **54** may comprise one or more imaging sensors of any suitable type(s), such as a charge-coupled device (CCD) sensor, a complementary metal-oxide-semiconductor (CMOS) sensor, an optical coherence tomography (OCT) sensor, and/or a hyperspectral image sensor. Using the sensors, the camera may acquire two-dimensional or three-dimensional images of any suitable type, such as monochrome images, color images (based, for example, on three color frames), multispectral images, hyperspectral images, optical coherence tomography (OCT) images, or images produced by fusing multiple images of different respective types.

(25) Typically, optical unit **30** further comprises a light source **66**, which is aligned, at least approximately, with path **92**. For example, the angle between path **92** and a hypothetical line extending from the end of path **92** on eye **25** to light source **66** may be less than 20 degrees, such as less than 10 degrees. Light source **66** is configured to function as a fixation target **64** by transmitting visible fixation light **68**, thus helping to stabilize the position of the eye. In particular, prior to the procedure, patient **22** is instructed to fixate eye **25** on light source **66**. Subsequently, during the procedure, by virtue of light source **66** transmitting fixation light **68**, eye **25** fixates on the light source, such that the eye's line-of-sight is approximately coincident with path **92** (due to the light source being approximately aligned with the path) and the eye is relatively stable. While the eye fixates on the light source, the radiation source irradiates the capsule of the eye with treatment beams **52**.

(26) In some embodiments, light source **66** comprises a light emitter, such as a light emitting diode (LED). In other embodiments, the light source comprises a reflector configured to reflect light emitted from a light emitter.

(27) Typically, the wavelength of fixation light **68**, which may be higher or lower than that of the treatment beams, is between 350 and 850 nm. For example, fixation light **68** may be orange or red, with a wavelength of 600-750 nm.

(28) Typically, the optical unit comprises an optical bench, and at least some of the aforementioned components belonging to the optical unit, such as the radiation source, the galvo mirrors, and the beam combiner, are coupled to the optical bench. Typically, the optical unit further comprises a front face **33**, through which the treatment beams and the fixation light pass. For example, optical unit **30** may comprise an encasement **31**, which at least partially encases the optical bench and comprises front face **33**. (Encasement **31** may be made of a plastic, a metal, and/or any other suitable material.) Alternatively, front face **33** may be attached to, or may be an integral part of, the optical bench.

(29) In some embodiments, front face **33** is shaped to define an opening **58**, through which the treatment beams and fixation light pass. In other embodiments, the front face comprises an exit window in lieu of opening **58**, such that fixation light **68** and treatment beams **52** pass through the exit window. The exit window may be made of a plastic, a glass, or any other suitable material that

is generally transparent to the treatment beams, aiming beams, fixation light **68**, and any light used for imaging the eye, such as light from the illumination sources **60** described below.

(30) Optical unit **30** is mounted onto an XYZ stage unit **32**, which is controlled by a control mechanism **36**, such as a joystick. Using control mechanism **36**, the user of system **20** may position the optical unit (e.g., by adjusting the distance of the optical unit from the eye) prior to treating the eye. In some embodiments, XYZ stage unit **32** comprises locking elements configured to inhibit motion of the stage unit following the positioning of the stage unit.

(31) In some embodiments, XYZ stage unit **32** comprises one or more motors **34**, and control mechanism **36** is connected to interface circuitry **46**. As the user manipulates the control mechanism, interface circuitry **46** translates this activity into appropriate electronic signals, and outputs these signals to controller **44**. In response to the signals, the controller controls motors **34**.

(32) In other embodiments, XYZ stage unit **32** is controlled manually by manipulating the control mechanism. In such embodiments, the XYZ stage unit may comprise a set of gears instead of motors **34**.

(33) System **20** further comprises a headrest **24**, comprising a forehead rest **26** and a chinrest **28**. During the capsulotomy, patient **22** presses his forehead against forehead rest **26** while resting his chin on chinrest **28**. In some embodiments, headrest **24** further comprises an immobilization strap **27**, configured to secure the patient's head from behind and thus keep the patient's head pressed against the headrest.

(34) In some embodiments, system **20** further comprises a contact lens, configured to focus the treatment beams and/or stabilize eye **25** while contacting the eye.

(35) In some embodiments, as shown in FIG. **1**, headrest **24** and XYZ stage unit **32** are both mounted onto a surface **38**, such as the top surface of a tray or table. In other embodiments, the XYZ stage unit is mounted onto surface **38**, and the headrest is attached to the XYZ stage unit.

(36) System **20** further comprises a monitor **42**, configured to display the images of the eye acquired by the camera. Monitor **42** may be attached to optical unit **30** or disposed at any other suitable location, such as on surface **38** next to device **21**. In some embodiments, monitor **42** comprises a touch screen, such that the user may input commands and/or information to the system via the touch screen. Alternatively or additionally, system **20** may comprise any other suitable input devices, such as a keyboard or a mouse, which may be used by the user.

(37) In some embodiments, monitor **42** is connected directly to controller **44** over a wired or wireless communication interface. In other embodiments, monitor **42** is connected to controller **44** via an external processor, such as a processor belonging to a standard desktop computer.

(38) In some embodiments, as shown in FIG. **2**, controller **44** is disposed within XYZ stage unit **32**. In other embodiments, controller **44** is disposed externally to the XYZ stage unit. Alternatively or additionally, the controller may cooperatively perform at least some of the functionality described herein with another, external processor.

(39) In some embodiments, system **20** is also configured to perform trabeculoplasty procedures, e.g., as described in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference. Thus, using system **20**, a trabeculoplasty procedure may be performed on patient **22** before or after a capsulotomy procedure is performed on the patient; for example, both procedures may be performed during a single sitting of patient **22**.

(40) In such embodiments, the treatment wavelength used for trabeculoplasty procedures (e.g., 532 nm) may be the same as that used for capsulotomy procedures. Alternatively, radiation source **48** may comprise a wavelength converter, configured to convert the wavelength of the treatment beams from a first wavelength used for capsulotomy procedures (e.g., 1064 nm) to a second wavelength used for trabeculoplasty procedures (e.g., 532 nm), or vice versa. Typically, the radiation source further comprises a switch, configured to switch the treatment beams between a first optical path, which includes the wavelength converter, and a second optical path, which does not include the wavelength converter.

(41) Alternatively or additionally, in such embodiments, optical unit **30** may further comprise one or more illumination sources **60** comprising, for example, one or more LEDs, such as white-light or infrared LEDs. For example, the optical unit may comprise a ring of LEDs surrounding opening **58**. In such embodiments, controller **44** may cause illumination sources **60** to intermittently flash light at the eye during a trabeculoplasty procedure, as described in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference. This flashing may facilitate the imaging performed by the camera, and, by virtue of the brightness of the flashing, may further help constrict the pupil of the eye. (For ease of illustration, the electrical connection between controller **44** and illumination sources **60** is not shown explicitly in FIG. 2.) In some embodiments, illumination sources **60** are coupled to front face **33**, as shown in FIG. 2.

(42) As shown in FIG. 1, in some embodiments—particularly those in which a trabeculoplasty, which generally requires that the entire limbus be visible, is performed—the optical unit is directed obliquely upward toward the eye while the eye gazes obliquely downward toward the optical unit, such that path **92** is oblique. For example, the path may be oriented at an angle θ of between five and twenty degrees with respect to the horizontal. Advantageously, this orientation reduces occlusion of the patient's eye by the patient's upper eyelid and associated anatomy.

(43) In some embodiments, as shown in FIG. 1, the oblique orientation of path **92** is achieved by virtue of the optical unit being mounted on a wedge **40**, which is mounted on the XYZ stage unit. In other words, the optical unit is mounted onto the XYZ stage unit via wedge **40**. (Wedge **40** is omitted from FIG. 2.)

(44) In some embodiments, at least some of the functionality of controller **44**, as described herein, is implemented in hardware, e.g., using one or more fixed-function or general-purpose integrated circuits, Application-Specific Integrated Circuits (ASICs), and/or Field-Programmable Gate Arrays (FPGAs). Alternatively or additionally, controller **44** may perform at least some of the functionality described herein by executing software and/or firmware code. For example, controller **44** may be embodied as a programmed processor comprising, for example, a central processing unit (CPU) and/or a Graphics Processing Unit (GPU). Program code, including software programs, and/or data may be loaded for execution and processing by the CPU and/or GPU. The program code and/or data may be downloaded to the controller in electronic form, over a network, for example. Alternatively or additionally, the program code and/or data may be provided and/or stored on non-transitory tangible media, such as magnetic, optical, or electronic memory. Such program code and/or data, when provided to the controller, produce a machine or special-purpose computer, configured to perform the tasks described herein.

(45) In some embodiments, the controller comprises a system on module (SOM), such as the DART-MX8M by Variscite of Lod, Israel.

Defining the Treatment Zone

(46) Reference is now made to FIG. 3, which is a schematic illustration of a method for defining a treatment zone, in accordance with some embodiments of the present invention.

(47) FIG. 3 shows eye **25** at the start of the capsulotomy procedure. In particular, FIG. 3 shows the anterior capsule **76** of the eye lying within the edge **78** of the iris **74** of the eye. An opening **84** has been formed in anterior capsule **76**, so as to allow the natural lens of the eye to be removed (e.g., via phacoemulsification) and a prosthetic intraocular lens (IOL) **80** to be inserted. Behind IOL **80** lies an opacified posterior capsule **86** of the eye, on which the capsulotomy procedure is to be performed.

(48) Typically, IOL **80** includes an elliptical optic **89** connected to curved filaments, or “haptics,” that anchor the IOL in place. For example, optic **89** may be circular, having a diameter of between 6 and 14 mm. Although IOL **80** is generally transparent, in some cases, as shown in FIG. 3, one or more features of IOL **80** (in particular, optic **89** of the IOL) may be visible behind opening **84**. For example, in some cases, optic **89** includes a Fresnel lens including multiple lens zones, and concentric circles **82**, which demarcate the lens zones, are visible.

(49) At the start of the capsulotomy procedure, controller **44** (FIG. 2) defines a treatment zone **88**, demarcated in the present figures by a boundary **90**, on posterior capsule **86**. Typically, treatment zone **88** is elliptical, e.g., circular.

(50) For example, based on at least one initial image of the eye acquired by camera **54** (FIG. 2), the controller may identify opening **84** in anterior capsule **76**, e.g., by applying any suitable pattern-matching and/or edge-detection techniques to the initial image. Subsequently, the controller may define the treatment zone such that the treatment zone lies entirely behind the opening. For example, the controller may offset boundary **90** a predefined distance (e.g., 0.1-1 mm) inward from the perimeter of opening **84**. As an extra safety precaution, the controller may further identify edge **78** of iris **74** based on the initial image, and verify that the treatment zone lies entirely within edge **78**.

(51) Alternatively, based on the initial image of the eye, the controller may identify one or more features of IOL **80**, e.g., by applying any suitable pattern-matching and/or edge-detection techniques to the initial image. In response to the features, the controller may calculate an estimated position of the IOL, and then define the treatment zone in response to the estimated position. In particular, the controller may define the position and size of the treatment zone such that, even if the entire treatment zone were to be opened, the IOL would not fall through the opening. As an extra safety precaution, the controller may further verify that the treatment zone lies entirely within edge **78**, and/or verify that the treatment zone lies entirely behind opening **84**.

(52) For example, prior to the procedure, a user may input, to the controller, relevant features of the IOL, such as the dimensions of, and/or number of lens zones belonging to, optic **89**. Subsequently, based on the input and in response to identifying one or more circles **82**, the controller may calculate the position of the center of the optic. In response thereto, the controller may center the treatment zone at the center of the optic, and/or offset boundary **90** a predefined distance (e.g., 0.5-1 mm) inward from the perimeter of the optic, e.g., per guidelines specified in the relevant medical literature.

(53) Alternatively, the controller may define the treatment zone by offsetting boundary **90** a predefined distance (e.g., up to 1 mm) inward from edge **78**, without necessarily identifying opening **84** or any features of IOL **80**.

(54) Typically, subsequently to defining the treatment zone, the controller overlays boundary **90**, which demarcates the treatment zone, on the initial image of the eye, and displays the image with the overlaid boundary on monitor **42** (FIG. 1). The user may then adjust the treatment zone, e.g., by using a mouse to drag one or more points on boundary **90**, prior to approving the treatment zone.

(55) In other embodiments, the controller defines the treatment zone in response to input from the user, without necessarily performing any of the image processing described above. For example, the user may input the desired center and radius of the treatment zone, and/or drag points on boundary **90** as described above.

Forming the Opening

(56) Subsequently to defining treatment zone **88**, the controller forms an opening in capsule **86** by irradiating multiple target regions **94** within the treatment zone in an iterative process. During each iteration of the process, the controller acquires an image of at least part of the capsule, designates one of the target regions **94** based on the acquired image, and causes radiation source **48** (FIG. 2) to irradiate the designated target region. For example, FIG. 3 shows an initial target region at the center of treatment zone **88**, which the controller may designate and then irradiate during the first iteration of the process.

(57) For further details, reference is now made to FIG. 4, which is a schematic illustration of the formation of an opening **96** in capsule **86**, in accordance with some embodiments of the present invention.

(58) Section A of FIG. 4 shows an image **98** of the eye following the initialization of opening **96**, which is performed by irradiating the first designated target region **94** shown in FIG. 3. Based on

the image, the controller designates a second target region **94**, e.g., as further described below with reference to FIG. 5.

(59) Section B of FIG. 4 shows another image **98** following the irradiation of the second target region, which expands opening **96**. Based on the image, the controller designates a third target region **94**.

(60) Section C of FIG. 4 shows another image **98** following a number of further iterations in which target regions **94** are irradiated. Based on such an image, the controller may decide to refrain from designating any further target regions, thus terminating the iterative process.

(61) For example, the controller may define a target boundary **91** for the opening, e.g., by placing target boundary **91** at a predefined offset (e.g., 0.01-1 mm) inward from boundary **90** of the treatment zone. Subsequently, the controller may terminate the process in response to ascertaining that the opening reaches a predefined threshold percentage (e.g., 90%-100%) of target boundary **91**. (In such embodiments, prior to the first treatment beam, target boundary **91** may be overlaid on the initial image of the eye and the user may then adjust the target boundary, as described above for boundary **90** with reference to FIG. 3.) Alternatively (e.g., if the pupil of the eye is relatively small), the controller may terminate the process in response to ascertaining that the opening reaches a predefined threshold percentage (e.g., 90%-100%) of boundary **90** of the treatment zone.

(62) Reference is now made to FIG. 5, which is a schematic illustration of a technique for designating target regions **94**, in accordance with some embodiments of the present invention.

(63) FIG. 5 shows the scenario generally shown in Sections A-B of FIG. 4, in which opening **96** is being formed. In this scenario, in some cases, the controller designates the next target region by performing the following technique.

(64) First, based on image **98**, the controller identifies a portion of the perimeter of the opening that is farther from boundary **90** than are other portions of the perimeter.

(65) For example, for multiple points **104a** along the perimeter, the controller may calculate the shortest distance **d0a** between point **104a** and boundary **90**, i.e., the length of the shortest line between point **104a** and boundary **90**. Subsequently, the controller may identify the point **104a** for which **d0a** is a maximum.

(66) Alternatively, for multiple angles, the controller may calculate the distance **d0b** between (i) the point **104b** at which a line **102** radiating from the center **100** of the treatment zone at the angle intersects the perimeter of opening **96**, and (ii) the point at which line **102** intersects boundary **90**. Subsequently, the controller may identify the point **104b** for which **d0b** is a maximum.

(67) Next, the controller designates target region **94** at a predefined distance **d1** from the identified portion of the perimeter. For example, subsequently to identifying a point **104a** as described above, the controller may designate a target region **94a** lying on the shortest line between the identified point and boundary **90** at a distance of **d1** from the identified point **104a**. Similarly, subsequently to identifying a point **104b** as described above, the controller may designate a target region **94b** lying on the line **102** joining the identified point to boundary **90**, at a distance **d1** from the identified point **104b**.

(68) Reference is now made to FIG. 6, which is a schematic illustration of another technique for designating target region **94**, in accordance with some embodiments of the present invention.

(69) In some cases, at least one fold **106** in the tissue of the capsule lies within the treatment zone. In such cases, the controller may identify fold **106**, e.g., by applying an edge-detection algorithm, a line-detection algorithm, or any other suitable image-processing algorithm to image **98**.

Subsequently, in response to identifying the fold, the controller may designate target region **94** such that the target region overlaps the fold.

(70) For example, for each of the identified folds, the controller may calculate two distances: (i) the distance **d2** between the center of the fold and the nearest point on the perimeter of the opening, and (ii) the distance **d3** between the center of the fold and the nearest point on boundary **90**. The controller may then identify a set of “targetable” folds for which **d2** is within a predefined range of

distances (e.g., 0.2-2 mm, such as 0.5-1 mm). Subsequently, the controller may select the targetable fold for which **d3** is a maximum, and then place target region **94** at the center of the selected fold. Alternatively, if there are no targetable folds, the controller may instead use the technique of FIG. 5 to designate the target region.

(71) Reference is now made to FIG. 7, which is a schematic illustration of a technique for designating target regions based on a predesignated sequence of tentative target regions, in accordance with some embodiments of the present invention.

(72) In some embodiments, prior to irradiating the capsule (and subsequently to the definition of the treatment zone), a sequence of tentative target regions **130** is designated. For example, the controller may display an image of the capsule to the user, and the user may then manually designate the sequence by clicking a mouse over each portion of the image at which a tentative target region **130** is desired. Alternatively, the controller may designate the sequence and display the sequence to the user, and the user may then approve the sequence after shifting, adding, or deleting any number of tentative target regions **130**.

(73) Typically, the sequence generally follows a geometric pattern. For example, the tentative target regions may be arranged in a spiral **132**, which spirals outward from the center of the treatment zone, or in multiple crossed lines that define an asterisk. Parts of the sequence may deviate from the pattern slightly, e.g., so as to allow one or more tentative target regions **130** to overlap folds **106** (FIG. 6).

(74) In such embodiments, when forming opening **96**, the controller iterates through the sequence of tentative target regions **130**. For each of the tentative target regions, the controller either (i) designates the tentative target region as the next target region **94**, i.e., approves the tentative target region for irradiation, as illustrated in the top-right inset portion of FIG. 7, (ii) designates the next target region by applying an offset (typically away from the perimeter of opening **96**) to the location of the tentative target region, as illustrated in the middle-right inset portion of FIG. 7, or (iii) designates the following tentative target region as the next target region, as illustrated in the bottom-right inset portion of FIG. 7.

(75) In particular, for each tentative target region, the controller ascertains whether the distance **d2** between the tentative target region and the perimeter of opening **96** is greater than a predefined threshold distance (e.g., 0.2-0.9 mm). If **d2** is greater than the predefined threshold distance, the tentative target region is designated as the target region **94**. Otherwise, the controller either applies the offset or designates the following tentative target region.

(76) Notwithstanding the examples shown in FIGS. 4-7, it is noted that opening **96** need not necessarily be initialized at the center of the treatment zone. For example, when sweeping across the capsule in a cross or asterisk pattern, the first target region in each line of the cross or asterisk may be designated near the edge of the treatment zone.

Example Iterative Process

(77) Reference is now made to FIG. 8, which is a flow diagram for an example iterative process **108** for forming an opening in a capsule of an eye, in accordance with some embodiments of the present invention. Iterative process **108** may be performed by the controller subsequently to defining the treatment zone, e.g., as described above with reference to FIG. 3.

(78) Each iteration of process **108** begins with an imaging step **110**, at which the controller, using camera **54** (FIG. 2), acquires an image of at least part of the capsule. Subsequently to imaging step **110**, the controller, at a location-identifying step **112**, identifies the location of boundary **90** (FIG. 3) of the treatment zone in the field of view (FOV) of the camera, by repeating whichever image processing was used previously to define the treatment zone. (The location of the treatment-zone boundary in the FOV may change throughout the procedure, due to movement of the eye.)

(79) For example, if the treatment zone was defined by placing boundary **90** at a predefined distance from the perimeter of opening **84** (FIG. 3), the controller may identify the perimeter of opening **84** in the latest acquired image, and then identify the location of the treatment-zone

boundary at the predefined distance therefrom. For embodiments in which target boundary **91** (FIG. **4**) is defined, the controller may also identify the location of boundary **91** in the FOV of the camera.

(80) Subsequently, at a checking step **114**, the controller checks whether at least one target region was already irradiated (i.e., whether at least one iteration was already performed). If not, the controller performs a target-designating step **124**, described below. Otherwise, the controller checks, at another checking step **116**, whether the last-designated (and irradiated) target region was opened. In other words, based on the acquired image, the controller checks whether tissue of the capsule is present at the last-designated target region.

(81) Typically, checking step **116** comprises two sub-steps. In the first sub-step, the controller identifies the location of the last target region based on the location of boundary **90**. For example, based on the location of boundary **90**, the controller may calculate any shift in the location of the center of the treatment zone, and then apply the same shift to the location of the last target region. Thus, for example, assuming that (i) the last target region was centered at (x_0, y_0) with the center of the treatment zone being at $(0, 0)$, and (ii) the center of the treatment zone is now at (dx, dy) , the controller may identify $(x_0 + dx, y_0 + dy)$ as the new location of the last target region. Subsequently, in the second sub-step, the controller computes the change in pixel value between the last target region in the current image and the last target region in the last image, and compares this change to a predefined threshold.

(82) If the last target region was not opened (i.e., if the controller identifies capsular tissue at the last target region), the controller, at a target-redesignating step **126**, redesignates the last target region. In some embodiments, the controller may also increase, by a predefined increment, the energy with which the target region is subsequently irradiated. (In such embodiments, typically, the energy may be increased multiple times by the predefined increment up to a predefined maximum energy, after which, if the target region is still not opened, process **108** is aborted.)

(83) Alternatively, if the last target region was opened, the controller, at a perimeter-identifying step **118**, identifies the perimeter of opening **96** (FIG. **4**), e.g., using any suitable edge-detection algorithm. Subsequently, at a size-checking step **120**, the controller checks whether the opening is large enough, e.g., by checking whether the opening reaches target boundary **91** as described above with reference to FIG. **4**. If yes, process **108** ends. Otherwise, the controller proceeds to a stability-checking step **122**.

(84) At stability-checking step **122**, the controller checks the stability of the perimeter of the opening by comparing the current location of the perimeter relative to the treatment-zone boundary, and shape of the perimeter, to the last location and shape. In response to ascertaining that the perimeter is stable, the controller proceeds to target-designating step **124**. Alternatively, in response to ascertaining that the perimeter is not stable—i.e., that the opening is still expanding as a result of previous irradiation—the controller refrains from designating a new target region, and returns to imaging step **110**.

(85) At target-designating step **124**, the controller designates a new target region. For example, the controller may first check for any targetable folds, as described above with reference to FIG. **6**. If a targetable fold is found, the controller may target the fold, as further described above with reference to FIG. **6**. Otherwise, the controller may designate a target region as described above with reference to FIG. **5** or FIG. **7**.

(86) Subsequently to performing target-designating step **124** or target-redesignating step **126**, the controller, at an aiming step **128**, aims aiming beams **53** (FIG. **2**) at the designated target region. Subsequently, the controller acquires another image of the capsule at imaging step **110**. The controller then checks, at a checking step **134**, whether the aiming beams overlap one another at the designated target region. If not, the controller, at a distance-adjusting step **136**, adjusts the distance of optical unit **30** from the eye, e.g., by communicating appropriate control signals to motors **34** (FIG. **2**). Alternatively or additionally, the controller may adjust the position of at least one optic in

the optical unit, such as galvo mirrors **50** or focusing module **72** (FIG. 2). The controller then acquires another image of the capsule, and repeats checking step **134**. In this manner, the distance of the optical unit from the eye, and/or the position of the optic, may be iteratively adjusted by the controller. Alternatively, the distance of the optical unit from the eye may be manually adjusted by the user.

(87) Upon the controller (or the user) ascertaining that the aiming beams overlap one another, the controller causes radiation source **48** (FIG. 2) to irradiate the designated target region, at an irradiating step **138**. The controller then begins another iteration of process **108**.

(88) In some embodiments, the aiming beams are shaped to define different respective portions of a predefined composite pattern, such that the predefined composite pattern is formed on the capsule only when the aiming beams overlap one another. Examples of such patterns are described, for example, in International Patent Application Publication WO/2020/008323, whose disclosure is incorporated herein by reference.

(89) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. A system, comprising: a radiation source; and a controller, configured to: define a treatment zone on a capsule of an eye of a subject, and subsequently to defining the treatment zone, form an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process: acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions on the at least part of the capsule, and causing the radiation source to irradiate the designated target region.
2. The system according to claim 1, wherein the capsule is a posterior capsule.
3. The system according to claim 2, wherein the controller is configured to define the treatment zone by: based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye, and defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.
4. The system according to claim 1, wherein the controller is configured to define the treatment zone by: based on at least one initial image of the eye, identifying an edge of an iris of the eye, and defining the treatment zone such that the treatment zone lies entirely within the edge.
5. The system according to claim 1, wherein the controller is configured to define the treatment zone by: based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye, in response to the features, calculating an estimated position of the IOL, and defining the treatment zone in response to the estimated position.
6. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and in response to identifying the tissue, redesignating the last-designated one of the target regions.
7. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and designating the target region at a predefined distance from the identified portion of the perimeter.
8. The system according to claim 1, wherein, during at least one of the iterations, designating the

target region includes: based on the image, identifying a fold within the treatment zone, and in response to identifying the fold, designating the target region such that the target region overlaps the fold.

9. The system according to claim 1, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, ascertaining that a perimeter of the opening is stable, and in response to the ascertaining, designating the target region.

10. The system according to claim 1, wherein the controller is configured to form the opening subsequently to a designation of a sequence of tentative target regions, and wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.

11. The system according to claim 10, wherein, during at least one other one of the iterations, designating the target region includes: based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.

12. The system according to claim 10, wherein, during at least one other one of the iterations, designating the target region includes: based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the target region by applying an offset to a location of the next one of the tentative target regions.

13. The system according to claim 1, wherein the controller is further configured to: define a target boundary on the capsule, and terminate the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.

14. The system according to claim 13, wherein the controller is configured to define the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.

15. A method, comprising: defining a treatment zone on a capsule of an eye of a subject; and subsequently to defining the treatment zone, forming an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process: acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions on the at least part of the capsule, and causing a radiation source to irradiate the designated target region.

16. The method according to claim 15, wherein the capsule is a posterior capsule.

17. The method according to claim 16, wherein defining the treatment zone comprises: based on at least one initial image of the eye, identifying an anterior-capsule opening in an anterior capsule of the eye; and defining the treatment zone such that the treatment zone lies entirely behind the anterior-capsule opening.

18. The method according to claim 15, wherein defining the treatment zone comprises: based on at least one initial image of the eye, identifying an edge of an iris of the eye; and defining the treatment zone such that the treatment zone lies entirely within the edge.

19. The method according to claim 15, wherein defining the treatment zone comprises: based on at least one initial image of the eye, identifying one or more features of a prosthetic intraocular lens (IOL) in the eye; in response to the features, calculating an estimated position of the IOL; and defining the treatment zone in response to the estimated position.

20. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, identifying tissue of the capsule at a last-designated one of the target regions, and in response to identifying the tissue, redesignating the

last-designated one of the target regions.

21. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, identifying a portion of a perimeter of the opening that is farther from a boundary of the treatment zone than are other portions of the perimeter, and designating the target region at a predefined distance from the identified portion of the perimeter.

22. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes: based on the image, identifying a fold within the treatment zone, and in response to identifying the fold, designating the target region such that the target region overlaps the fold.

23. The method according to claim 15, wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, ascertaining that a perimeter of the opening is stable, and in response to the ascertaining, designating the target region.

24. The method according to claim 15, wherein forming the opening comprises forming the opening subsequently to a designation of a sequence of tentative target regions, and wherein, during at least one of the iterations, designating the target region includes: based on the acquired image, ascertaining that a distance between a next one of the tentative target regions and a perimeter of the opening is greater than a predefined threshold distance, and in response to ascertaining that the distance is greater than the predefined threshold distance, designating, as the target region, the next one of the tentative target regions.

25. The method according to claim 24, wherein, during at least one other one of the iterations, designating the target region includes: based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and in response to ascertaining that the distance is not greater than the predefined threshold distance, designating, as the target region, the tentative target region following the next one of the tentative target regions.

26. The method according to claim 24, wherein, during at least one other one of the iterations, designating the target region includes: based on the acquired image, ascertaining that the distance is not greater than the predefined threshold distance, and in response to ascertaining that the distance is not greater than the predefined threshold distance, designating the target region by applying an offset to a location of the next one of the tentative target regions.

27. The method according to claim 15, further comprising: defining a target boundary on the capsule; and terminating the iterative process in response to ascertaining that the opening reaches a predefined threshold percentage of the target boundary.

28. The method according to claim 27, wherein defining the target boundary comprises defining the target boundary by placing the target boundary at a predefined offset inward from a boundary of the treatment zone.

29. A computer software product comprising a tangible non-transitory computer-readable medium in which program instructions are stored, which instructions, when read by a controller, cause the controller to: define a treatment zone on a capsule of an eye of a subject, and subsequently to defining the treatment zone, form an opening in the capsule by irradiating multiple target regions within the treatment zone in an iterative process that includes, during each one of multiple iterations of the process: acquiring an image of at least part of the capsule, based on the acquired image, designating one of the target regions on the at least part of the capsule, and causing a radiation source to irradiate the designated target region.
